

ERASMUS UNIVERSITY ROTTERDAM

ERASMUS SCHOOL OF ECONOMICS

Master's Thesis Econometrics and Management Science - Business Analytics and Quantitative
Marketing

Median-of-Means Two-Stage Least Squares Regression

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Abstract

Median-of-Means methods are used for instrumental variable regression and their properties are compared to the standard 2-stage least squares method.

1 Preliminaries

1.1 Setup and Assumptions

Let $(Y_i, X_i, Z_i)_{i=1}^n$ be an i.i.d. random sample from a joint distribution on $\mathbb{R}^3$ satisfying the structural equation

$$Y_i = \beta X_i + \varepsilon_i, \quad (1)$$

where $\beta \in \mathbb{R}$ is the structural parameter of interest and ε_i is an unobserved error term. We impose the following conditions.

(A1) **Exogeneity.** $\mathbb{E}[Z_i \varepsilon_i] = 0$.

(A2) **Relevance.** $\mu_{ZX} := \mathbb{E}[Z_i X_i] \neq 0$.

(A3) **Finite second moments.** $\sigma_{ZY}^2 := \text{Var}(Z_i Y_i) < \infty$, $\sigma_{ZX}^2 := \text{Var}(Z_i X_i) < \infty$, and $\sigma_{Z\varepsilon}^2 := \text{Var}(Z_i \varepsilon_i) < \infty$.

Define $\mu_{ZY} := \mathbb{E}[Z_i Y_i]$. By (A1) and (A2), the structural parameter is identified as $\beta = \mu_{ZY} / \mu_{ZX}$.

Remark 1.1 (Moment conditions). *The conditions (A1)–(A3) hold for any distribution with finite second moments of the products $Z_i Y_i$, $Z_i X_i$, and $Z_i \varepsilon_i$. Thus, no assumptions are imposed on the marginal distributions of Y_i , X_i , Z_i , or ε_i .*

Remark 1.2 (Relationship between variances). *The three variances in (A3) are related. Since $Z_i Y_i = \beta Z_i X_i + Z_i \varepsilon_i$, Minkowski's inequality gives*

$$\sigma_{Z\varepsilon} \leq \sigma_{ZY} + |\beta| \sigma_{ZX}. \quad (2)$$

To see this, set $U = Z_i Y_i$ and $V = \beta Z_i X_i$, so that $U - V = Z_i \varepsilon_i$. By Minkowski's inequality for $p = 2$,

$$\sqrt{\text{Var}(U - V)} \leq \sqrt{\text{Var}(U)} + \sqrt{\text{Var}(V)},$$

which gives $\sigma_{Z\varepsilon} \leq \sigma_{ZY} + |\beta| \sigma_{ZX}$.

1.2 Sub-Gaussian Estimators

Definition 1.3 (Sub-Gaussian Estimators). *Let $(X_i)_{i=1}^n$ be a random sample with $\mathbb{E}[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2$, and let $\bar{\mu}(X)$ denote an estimator for μ . For some constant $L > 0$, $\bar{\mu}(X)$ is said to be a sub-Gaussian estimator for μ if*

$$\mathbb{P} \left[|\bar{\mu}(X) - \mu| > L\sigma \sqrt{\frac{\ln(1/\delta)}{n}} \right] \leq \delta \quad \text{for all } \delta \in (0, 1). \quad (3)$$

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1.3 The Median-of-Means Estimator for a Scalar Mean

We present the scalar Median-of-Means (MoM) estimator and its concentration properties. The results in this section are adapted from Devroye, Lerasle, Lugosi and Oliveira (2016) and Lugosi and Mendelson (2019).

Algorithm 1: Scalar Median-of-Means Estimator

Input: Sample $(X_i)_{i=1}^n$, number of blocks k
Output: Estimate $\hat{\mu}$
 Partition $(X_i)_{i=1}^n$ into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$;
for $j = 1, \dots, k$ **do**
 Compute the block mean $\bar{\mu}_j = \frac{1}{m} \sum_{i \in B_j} X_i$;
return $\hat{\mu} = \text{med}(\bar{\mu}_1, \dots, \bar{\mu}_k)$;

Theorem 1.4 (MoM concentration). *Let $(X_i)_{i=1}^n$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. Let k be a positive integer, $m = \lfloor n/k \rfloor$, and $\hat{\mu}_{\text{MoM}}$ the MoM estimator with k blocks. Then*

$$\mathbb{P} \left[|\hat{\mu}_{\text{MoM}} - \mu| > \frac{2\sigma}{\sqrt{m}} \right] \leq e^{-k/8}. \quad (4)$$

Setting $k = \lceil 8 \ln(1/\delta) \rceil$ for $\delta \in (0, 1)$, this becomes

$$\mathbb{P} \left[|\hat{\mu}_{\text{MoM}} - \mu| > \sigma \sqrt{\frac{32 \ln(1/\delta)}{n}} \right] \leq \delta, \quad (5)$$

which is sub-Gaussian in the sense that the dependence on δ is logarithmic.

Proof. The proof proceeds in three steps.

Step 1 (Chebyshev on each block). Each block mean satisfies $\mathbb{E}[\bar{X}_j] = \mu$ and $\text{Var}(\bar{X}_j) = \sigma^2/m$. By Chebyshev's inequality with threshold $2\sigma/\sqrt{m}$,

$$\mathbb{P} \left[|\bar{X}_j - \mu| > \frac{2\sigma}{\sqrt{m}} \right] \leq \frac{\sigma^2/m}{4\sigma^2/m} = \frac{1}{4}.$$

Step 2 (Counting argument for the median). Define $\zeta_j = \mathbf{1}\{|\bar{X}_j - \mu| > 2\sigma/\sqrt{m}\}$. By Step 1, $\mathbb{E}[\zeta_j] \leq 1/4$. Since the blocks are disjoint subsets of an i.i.d. sample, the $(\zeta_j)_{j=1}^k$ are independent.

If $|\hat{\mu}_{\text{MoM}} - \mu| > 2\sigma/\sqrt{m}$, the median exceeds $\mu + 2\sigma/\sqrt{m}$ or falls below $\mu - 2\sigma/\sqrt{m}$. In either case, at least $k/2$ of the block means satisfy $|\bar{X}_j - \mu| > 2\sigma/\sqrt{m}$, since the median of k numbers can only lie outside an interval if fewer than half the numbers lie inside it. Therefore,

$$\mathbb{P} \left[|\hat{\mu}_{\text{MoM}} - \mu| > \frac{2\sigma}{\sqrt{m}} \right] \leq \mathbb{P} \left[\sum_{j=1}^k \zeta_j \geq \frac{k}{2} \right].$$

Step 3 (Hoeffding's inequality). The sum $\sum_{j=1}^k \zeta_j$ consists of k independent random variables with $\zeta_j \in [0, 1]$ and $\mathbb{E}[\sum_j \zeta_j] \leq k/4$. For $\sum_j \zeta_j$ to reach $k/2$, it must exceed its mean by

at least $k/2 - k/4 = k/4$. By Hoeffding's inequality (Hoeffding, 1963),

$$\mathbb{P} \left[\sum_{j=1}^k \zeta_j - \mathbb{E} \left[\sum_j \zeta_j \right] \geq \frac{k}{4} \right] \leq \exp \left(\frac{-2(k/4)^2}{k} \right) = e^{-k/8}.$$

This establishes (4). The sub-Gaussian form (5) follows by setting $k = \lceil 8 \ln(1/\delta) \rceil$ and noting $m \geq n/(2k)$ for $n \geq 2k$. $\square$

2 The Standard Instrumental Variable Estimator

Definition 2.1 (Standard IV estimator). *Consider the setup given in 1. The standard IV estimator $\hat{\beta}_{\text{IV}}$ is given by*

$$\hat{\beta}_{\text{IV}} = \frac{\bar{S}_{ZY}}{\bar{S}_{ZX}}, \quad \bar{S}_{ZY} = \frac{1}{n} \sum_{i=1}^n Z_i Y_i, \quad \bar{S}_{ZX} = \frac{1}{n} \sum_{i=1}^n Z_i X_i.$$

Lemma 2.2 (Error decomposition). *Under (1) and (A1),*

$$\hat{\beta}_{\text{IV}} - \beta = \frac{\bar{S}_{Z\varepsilon}}{\bar{S}_{ZX}}, \tag{6}$$

where $\bar{S}_{Z\varepsilon} = \frac{1}{n} \sum_{i=1}^n Z_i \varepsilon_i$ satisfies $\mathbb{E}[\bar{S}_{Z\varepsilon}] = 0$.

Proof. Multiply both sides of $Y_i = \beta X_i + \varepsilon_i$ by Z_i and average:

$$\bar{S}_{ZY} = \frac{1}{n} \sum_{i=1}^n Z_i (\beta X_i + \varepsilon_i) = \beta \bar{S}_{ZX} + \bar{S}_{Z\varepsilon}.$$

Dividing by $\bar{S}_{ZX}$ gives $\hat{\beta}_{\text{IV}} = \beta + \bar{S}_{Z\varepsilon}/\bar{S}_{ZX}$. Furthermore, by (A1), $\mathbb{E}[Z_i \varepsilon_i] = 0$, so $\mathbb{E}[\bar{S}_{Z\varepsilon}] = 0$ by linearity of expectation. So $\hat{\beta}_{\text{IV}}$ is unbiased. $\square$

Theorem 2.3 (Standard IV concentration). *Assume (A1)–(A3). For any $\delta \in (0, 1)$, if*

$$n \geq \frac{8\sigma_{ZX}^2}{\delta \mu_{ZX}^2}, \tag{7}$$

then

$$\mathbb{P} \left[\left| \hat{\beta}_{\text{IV}} - \beta \right| > \frac{2\sqrt{2} \sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{\delta n}} \right] \leq \delta. \tag{8}$$

Proof. The proof uses the two-event strategy applied to the full-sample decomposition (6).

Step 1 (Two-event decomposition). For a threshold $t > 0$ to be determined, define

$$A = \left\{ |\bar{S}_{ZX} - \mu_{ZX}| \leq \frac{|\mu_{ZX}|}{2} \right\}, \quad B = \left\{ |\bar{S}_{Z\varepsilon}| \leq \frac{t |\mu_{ZX}|}{2} \right\}.$$

On event A , the reverse triangle inequality gives $|\bar{S}_{ZX}| \geq |\mu_{ZX}| - |\bar{S}_{ZX} - \mu_{ZX}| \geq |\mu_{ZX}|/2$.

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On $A \cap B$,

$$\left| \hat{\beta}_{\text{IV}} - \beta \right| = \frac{|\bar{S}_{Z\varepsilon}|}{|\bar{S}_{ZX}|} \leq \frac{t |\mu_{ZX}|/2}{|\mu_{ZX}|/2} = t.$$

Step 2 (Chebyshev on each event). Since $\text{Var}(\bar{S}_{ZX}) = \sigma_{ZX}^2/n$ and $\text{Var}(\bar{S}_{Z\varepsilon}) = \sigma_{Z\varepsilon}^2/n$, Chebyshev's inequality gives

$$\mathbb{P}[A^c] \leq \frac{\sigma_{ZX}^2/n}{\mu_{ZX}^2/4} = \frac{4\sigma_{ZX}^2}{n\mu_{ZX}^2}, \quad \mathbb{P}[B^c] \leq \frac{\sigma_{Z\varepsilon}^2/n}{t^2\mu_{ZX}^2/4} = \frac{4\sigma_{Z\varepsilon}^2}{nt^2\mu_{ZX}^2}.$$

Step 3 (Union bound and solve for t). We allocate $\delta/2$ to each event. The condition (7) ensures $\mathbb{P}[A^c] \leq \delta/2$. Setting $\mathbb{P}[B^c] \leq \delta/2$ and solving for t :

$$\frac{4\sigma_{Z\varepsilon}^2}{nt^2\mu_{ZX}^2} = \frac{\delta}{2} \implies t = \frac{2\sqrt{2}\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}}.$$

By the union bound, $\mathbb{P}\left[\left|\hat{\beta}_{\text{IV}} - \beta\right| > t\right] \leq \mathbb{P}[A^c] + \mathbb{P}[B^c] \leq \delta/2 + \delta/2 = \delta$. □

Remark 2.4 (Polynomial dependence on δ). *The bound (8) depends on δ through the factor $1/\sqrt{\delta}$, which is polynomial. Devroye et al. (2016) show that for any estimator based solely on the empirical mean of i.i.d. observations with finite variance, polynomial dependence on δ is unavoidable without additional distributional assumptions.*

Remark 2.5 (Instrument strength depends on δ). *The condition (7) involves δ : achieving higher confidence (smaller δ) requires more samples to control the denominator. This dependence is polynomial in $1/\delta$.*

3 The Ratio-of-Medians MoM-IV Estimator

Algorithm 2: Ratio-of-Medians Estimator

Input: Sample $(Y_i, X_i, Z_i)_{i=1}^n$, number of blocks k

Output: Estimate $\hat{\beta}_{\text{RM}}$

Partition $(Y_i, X_i, Z_i)_{i=1}^n$ into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$;

for $j = 1, \dots, k$ **do**

Compute the block means $\hat{S}_{ZY}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i Y_i$ and $\hat{S}_{ZX}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i X_i$;
 Compute coordinate-wise medians $\tilde{S}_{ZY} = \text{med}(\hat{S}_{ZY}^{(1)}, \dots, \hat{S}_{ZY}^{(k)})$ and
 $\tilde{S}_{ZX} = \text{med}(\hat{S}_{ZX}^{(1)}, \dots, \hat{S}_{ZX}^{(k)})$;

return $\hat{\mu} = \text{med}(\tilde{\mu}_1, \dots, \tilde{\mu}_k)$;

Theorem 3.1 (RoM concentration). *Assume (A1)–(A3). Let $\delta \in (0, 1)$, $k = \lceil 8 \ln(2/\delta) \rceil$, and $m = \lfloor n/k \rfloor$. If the instrument strength condition*

$$m > \frac{4\sigma_{ZX}^2}{\mu_{ZX}^2} \tag{9}$$

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holds, then

$$\mathbb{P}\left[\left|\hat{\beta}_{\text{RM}} - \beta\right| > \frac{2(\sigma_{ZY} + |\beta|\sigma_{ZX})}{|\mu_{ZX}|\sqrt{m} - 2\sigma_{ZX}}\right] \leq \delta. \quad (10)$$

Proof. The proof applies the scalar MoM result (Theorem 1.4) to each coordinate separately and combines the results with a union bound.

Step 1 (MoM on each coordinate). By Theorem 1.4 applied to the sequences $(Z_i Y_i)_{i=1}^n$ and $(Z_i X_i)_{i=1}^n$,

$$\mathbb{P}\left[\left|\tilde{S}_{ZY} - \mu_{ZY}\right| > \frac{2\sigma_{ZY}}{\sqrt{m}}\right] \leq e^{-k/8}, \quad \mathbb{P}\left[\left|\tilde{S}_{ZX} - \mu_{ZX}\right| > \frac{2\sigma_{ZX}}{\sqrt{m}}\right] \leq e^{-k/8}.$$

Step 2 (Union bound). Define the events

$$A = \left\{\left|\tilde{S}_{ZY} - \mu_{ZY}\right| \leq \frac{2\sigma_{ZY}}{\sqrt{m}}\right\}, \quad B = \left\{\left|\tilde{S}_{ZX} - \mu_{ZX}\right| \leq \frac{2\sigma_{ZX}}{\sqrt{m}}\right\}.$$

By Step 1, $\mathbb{P}[A^c] \leq e^{-k/8}$ and $\mathbb{P}[B^c] \leq e^{-k/8}$. Note that A and B are not independent (both are constructed from the same data), so we use the union bound:

$$\mathbb{P}[(A \cap B)^c] = \mathbb{P}[A^c \cup B^c] \leq \mathbb{P}[A^c] + \mathbb{P}[B^c] \leq 2e^{-k/8}.$$

Setting $k = \lceil 8 \ln(2/\delta) \rceil$ ensures $2e^{-k/8} \leq \delta$.

Step 3 (Algebraic bound on the ratio). On $A \cap B$, we decompose

$$\hat{\beta}_{\text{RM}} - \beta = \frac{\tilde{S}_{ZY}}{\tilde{S}_{ZX}} - \frac{\mu_{ZY}}{\mu_{ZX}} = \frac{\tilde{S}_{ZY}\mu_{ZX} - \mu_{ZY}\tilde{S}_{ZX}}{\tilde{S}_{ZX} \cdot \mu_{ZX}}.$$

Adding and subtracting $\mu_{ZY}\mu_{ZX}$ in the numerator and using $\mu_{ZY} = \beta\mu_{ZX}$,

$$\tilde{S}_{ZY}\mu_{ZX} - \mu_{ZY}\tilde{S}_{ZX} = \mu_{ZX} \left[(\tilde{S}_{ZY} - \mu_{ZY}) - \beta(\tilde{S}_{ZX} - \mu_{ZX}) \right].$$

The factor μ_{ZX} cancels with the μ_{ZX} in the denominator, giving

$$\hat{\beta}_{\text{RM}} - \beta = \frac{(\tilde{S}_{ZY} - \mu_{ZY}) - \beta(\tilde{S}_{ZX} - \mu_{ZX})}{\tilde{S}_{ZX}}.$$

Taking absolute values and applying the triangle inequality to the numerator,

$$\left|\hat{\beta}_{\text{RM}} - \beta\right| \leq \frac{\left|\tilde{S}_{ZY} - \mu_{ZY}\right| + |\beta| \left|\tilde{S}_{ZX} - \mu_{ZX}\right|}{\left|\tilde{S}_{ZX}\right|}.$$

Step 4 (Substitute bounds). On event A , $\left|\tilde{S}_{ZY} - \mu_{ZY}\right| \leq 2\sigma_{ZY}/\sqrt{m}$. On event B , $\left|\tilde{S}_{ZX} - \mu_{ZX}\right| \leq 2\sigma_{ZX}/\sqrt{m}$, and by the reverse triangle inequality,

$$\left|\tilde{S}_{ZX}\right| \geq |\mu_{ZX}| - \left|\tilde{S}_{ZX} - \mu_{ZX}\right| \geq |\mu_{ZX}| - \frac{2\sigma_{ZX}}{\sqrt{m}}.$$

The instrument strength condition (9) ensures $2\sigma_{ZX}/\sqrt{m} < |\mu_{ZX}|$, so the denominator is strictly positive. Substituting,

$$\left| \hat{\beta}_{\text{RM}} - \beta \right| \leq \frac{2\sigma_{ZY}/\sqrt{m} + |\beta| \cdot 2\sigma_{ZX}/\sqrt{m}}{|\mu_{ZX}| - 2\sigma_{ZX}/\sqrt{m}} = \frac{2(\sigma_{ZY} + |\beta| \sigma_{ZX})}{|\mu_{ZX}| \sqrt{m} - 2\sigma_{ZX}}.$$

Since this holds on $A \cap B$ and $\mathbb{P}[(A \cap B)^c] \leq \delta$, the result follows. $\square$

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